CARLOS GADIEL PERALTA PEÑA

Junior Backend Developer | Python · FastAPI · SQL · Databases

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PROFESSIONAL SUMMARY

Computer Systems Engineer (degree and professional license) focused on databases and backend development with Python. Built an end-to-end tire crack detection system (computer vision model, FastAPI REST API and mobile client app) and designed relational databases that cut query times from minutes to seconds. Self-taught, analytical and results-driven. English B2. Seeking a Junior Backend / Database Developer role (on-site, hybrid or remote). Available immediately.

TECHNICAL SKILLS

Main language: Python
Backend & APIs: FastAPI · REST APIs · Endpoint documentation · Git & GitHub
Databases: MySQL · PostgreSQL · SQL · Entity-relationship modeling · Schema design
Data: Pandas · NumPy · ETL · Automated reporting · Advanced Excel
Computer vision: OpenCV · Image processing · Segmentation · CNNs · Deep Learning
Working knowledge: Java · Kotlin · C++ · JavaScript · HTML · CSS · Linux (basic) · Mobile apps
Hardware & support: Computer maintenance and troubleshooting · Custom PC building

MAIN PROJECT

Car Tire Crack Detection | Degree project · ITSM · 2025 – Feb 2026
Stack: Python · FastAPI · OpenCV · NumPy · Deep Learning · REST API | Screenshots: gaxtec.github.io/portafolio/#proyectos

- Designed and built a REST API with FastAPI that receives images from a mobile app and classifies in real time whether a tire is cracked.
- Implemented a computer vision pipeline that, on a positive detection, segments the image with masks differentiated by type of damage.
- Built the full flow: mobile app, API, model inference and a response with the segmented image overlaid.
- Trained and compared three architectures (DenseNet121, VGG19-FC6 and VGG19-FC7) on an industrial image dataset with data augmentation; the best model reached 90.7% accuracy and 0.90–0.91 F1 per class on 162 evaluation images.
- Documented the API endpoints and the entity-relationship model of the analysis logging system.

PROFESSIONAL EXPERIENCE

Independent IT & Data Technician | Self-employed · 2020 – 2022

- Automated inventory and customer-tracking reports with advanced Excel, eliminating ~3 hours of manual work per week.
- Designed and implemented a relational database for service and customer records, cutting history lookups from minutes to seconds.
- Diagnosed and fixed faults on 50+ machines with a repeatable process that reduced rework by 60%.

OTHER PROJECTS

Portfolio website with contact API | 2026

- Live site with a FastAPI backend for its contact form: input validation and sanitization, per-IP rate limiting, anti-spam protection and email delivery via API.

Database Management System | ITSM 2023

- Designed and implemented a relational database to manage academic records and documented the entity-relationship model and functional requirements of the system.

Mobile App with REST API | ITSM 2024

- Built a working app from requirements analysis to deployment, with data storage and API consumption; managed design, development, testing and documentation.

EDUCATION

B.Eng. in Computer Systems Engineering | Instituto Tecnológico Superior de Misantla (ITSM) · 2021 – Feb 2025

- Degree conferred: Feb 2026 (professional degree and license, título y cédula profesional). Coursework in programming, data structures, databases, systems analysis and software development. Thesis: Car tire crack detection.

LANGUAGES

Spanish: native · **English (B2):** technical reading, professional written and oral communication